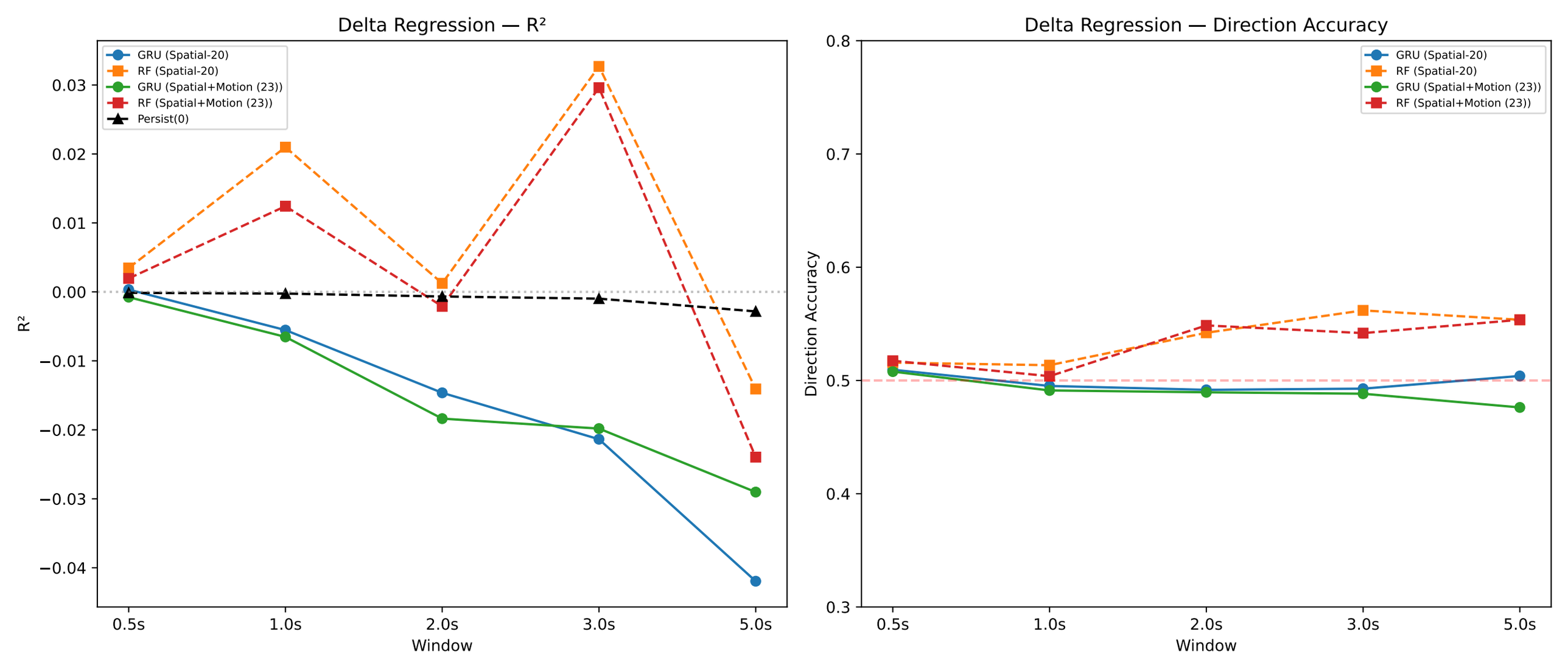
Delta Regression — R²
GRU (Spatial-20)
RF (Spatial-20)
GRU (Spatial+Motion (23))
RF (Spatial+Motion (23))
Persist(0)
R²
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
0.5s
1.0s
2.0s
3.0s
5.0s
Window
Delta Regression — Direction Accuracy
Direction Accuracy
0.8
0.7
0.6
0.5
0.4
0.3
GRU (Spatial-20)
RF (Spatial-20)
GRU (Spatial+Motion (23))
RF (Spatial+Motion (23))
0.5s
1.0s
2.0s
3.0s
5.0s
Window

Delta Regression: Predict CHANGE in difficulty
Target: mean(fn_nano next W) - mean(fn_nano current W)
Persistence baseline: always predict delta=0

Window	Features	GRU R²	RF R²	Persist R²	GRU Dir	RF Dir
0.5s	Spatial-20	0.0003	0.0035	-0.0001	0.509	0.516
0.5s	Spatial+Motion (23)	-0.0008	0.0020	-0.0001	0.508	0.517
1.0s	Spatial-20	-0.0055	0.0210	-0.0002	0.495	0.513
1.0s	Spatial+Motion (23)	-0.0065	0.0124	-0.0002	0.491	0.504
2.0s	Spatial-20	-0.0146	0.0012	-0.0007	0.492	0.542
2.0s	Spatial+Motion (23)	-0.0184	-0.0021	-0.0007	0.490	0.549
3.0s	Spatial-20	-0.0214	0.0327	-0.0010	0.493	0.562
3.0s	Spatial+Motion (23)	-0.0198	0.0296	-0.0010	0.488	0.542
5.0s	Spatial-20	-0.0420	-0.0141	-0.0028	0.504	0.554
5.0s	Spatial+Motion (23)	-0.0290	-0.0240	-0.0028	0.476	0.554